THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance of the point $(-5, 12)$ from the origin is: 1

(a) 14 units **(b)** 17 units **(c)** 13 units **(d)** 5 units

2. The sequence $1, 7, 13, 19, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $6n - 5$ **(b)** $6n + 1$ **(c)** $6n - 7$ **(d)** $n + 6$

3. The block 142857 repeats in the decimal of: 1

(a) $\frac{1}{9}$ **(b)** $\frac{1}{11}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{7}$

4. $(x + 7)(x + 8)$ equals: 1

(a) $x^2 + 15x - 56$ **(b)** $x^2 - x + 56$ **(c)** $x^2 + 56x + 15$ **(d)** $x^2 + 15x + 56$

5. In a circle of radius 10 cm, a chord is 8 cm from the centre. The length of the chord is: 1

(a) 12 cm **(b)** 6 cm **(c)** 14 cm **(d)** 10 cm

6. By Heron's formula, the area of a triangle with sides 3 cm, 4 cm and 5 cm is: 1

(a) 18 cm^2 **(b)** 6 cm^2 **(c)** 4 cm^2 **(d)** 12 cm^2

7. An event occurred 46 times in 200 trials. Its experimental probability (relative frequency) is: 1

(a) 0.23 **(b)** 4.35 **(c)** 0.33 **(d)** 0.115

8. A bag contains 9 red, 3 blue and 3 green balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{4}{5}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{1}{2}$

9. The 12th term of the AP $-5, -1, 3, \ldots$ is: 1

(a) 40 **(b)** 39 **(c)** 38 **(d)** 78

10. Which of the following is a counterexample to the proposition "If a number is divisible by 3, then it is divisible by 9."? 1

(a) 45 **(b)** 12 **(c)** 9 **(d)** 27

11. If a proposition is true, its converse: 1

(a) is always false **(b)** may be true or false

(c) is always true **(d)** is never a proposition

12. A house used 150, 450, 250 and 150 units of electricity on lighting, cooling, kitchen and other appliances in 2018. In a 100% stacked bar for 2018, the length of the cooling segment is: 1

(a) 90% **(b)** 45% **(c)** 50% **(d)** 40%

13. 600 mL of a solution with 8% salt is mixed with 400 mL of a solution with 13% salt. The concentration of salt in the mixture is: 1

(a) 12% **(b)** 11% **(c)** 10% **(d)** 9%

14. If Step 5 (writing the final carry) is skipped, then 856 + 479 would be written as: 1

(a) 1335 **(b)** 335 **(c)** 1235 **(d)** 3350

15. A quadrilateral is convex when: 1

(a) one internal angle is more than 180°
(b) all its internal angles are less than 180°
(c) its diagonals do not intersect
(d) its sides are all equal

16. G is the centroid of △ABC and GD = 7 cm, where AD is a median. Then AD is: 1

(a) 28 cm **(b)** 21 cm **(c)** 14 cm **(d)** 22 cm

17. The digits of a two-digit number add up to 11 and the tens digit exceeds the units digit by 3. The number is: 1

(a) 74 **(b)** 83 **(c)** 65 **(d)** 47

18. **Assertion (A):** The slope of the line x = 3 is 0. 1
Reason (R): For the line x = 3 the run between any two points is 0, so the slope is undefined.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

19. Taking $\pi = \frac{22}{7}$, the curved surface area of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 462 cm^2 **(b)** 1540 cm^2 **(c)** 220 cm^2 **(d)** 440 cm^2

20. **Assertion (A):** A cube and a cuboid of equal volume need not have equal total surface areas. 1
Reason (R): Total surface area depends on the individual dimensions, not only on the product lwh.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. The temperature in Ladakh is 4 °C at noon and drops by 15 °C by midnight. Find the midnight temperature. 2

22. A circle has radius 13 cm. Two chords of length 10 cm each are drawn. Find their distance from the centre and state the relation between the distances. 2

23. Find the perimeter of a quadrant (quarter-circle) of radius 14 cm, including the two radii. ($\pi = \frac{22}{7}$) 2

24. If the probability that a team wins is 0.35, what is the probability that it does not win? Why? 2

25. Find a counterexample to "If n is an even number, then $2^n + 1$ is a prime number". 2

26. Three angles of a quadrilateral are $85°$, $95°$ and $110°$. Find the fourth angle. 2

27. A ramp rises 3 m over a horizontal run of 12 m. Find its slope and state what it means. 2

28. A cone and a cylinder have the same base radius and the same height. Find the ratio of their volumes. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 40–60 words each.

29. Divisors occur in pairs. Write the divisors of 36 in pairs and state how many numbers need to be examined to find all divisors of 36. 3

30. Sarita has a 500-page book and reads 20 pages a day. Write the pages left after d days and find the days needed to finish. 3

31. The radii of two spheres are in the ratio $3 : 5$. Find the ratio of their 3
(i) surface areas,
(ii) volumes.

32. A tailor has 15 m of silk and a kurta needs 2 m. How many kurtas can be made and how much silk is left? Express the number of kurtas as a rational number. 3

33. A rectangle has perimeter 36 cm. Find the dimensions of the rectangle with the greatest area and that area. 3

34. A sticks-and-diagonals model: a quadrilateral is of type Q when it has equal diagonals. 3
(a) Must the two sticks have equal length?
(b) Does the way they are placed matter if the diagonals only need to be equal?

35. Given $A(2, -2)$ and $B(8, 4)$, find the point P on AB that divides it in the ratio $1 : 2$ (nearer to A) and verify that $AP = \frac{1}{3}AB$. 3

OR

Show that $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$ are the vertices of an isosceles right-angled triangle. 3

36. A number is chosen at random from 1 to 25. Find the probability that it is 3
(i) a multiple of 5
(ii) a prime number.

OR

One month is chosen at random. Find the probability that it has exactly 30 days. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** The graph of the line l passes through $A(0, -2)$ and $B(4, 6)$. 5
(i) Find its slope.
(ii) Find the equation of l.
(iii) Where does l meet the x-axis?
(iv) A second line $x + y = 4$ is drawn on the same graph. Find the point where the two lines meet.

OR

Case study: A tank is filled so that the water level y (in cm) after x minutes follows the line $y = 4x$. 5

(i) What is the slope and what does it mean?

(ii) How high is the water after 6 minutes?

(iii) After how many minutes does it reach 50 cm?

(iv) Does the point $(8, 30)$ lie on this line? Justify.

(v) The tank's inlet is changed so that the level is $y = 4x + 10$. What does the 10 represent?

38. **Case study:** A triangular garden ABC has medians AD, BE and CF meeting at G. AD $= 27$ m, BE $= 36$ m. 5

(i) State the ratio in which G divides each median.

(ii) Find AG, GD, BG, GE.

(iii) If the third median CF $= 30$ m, find CG and GF.

(iv) Why are the three medians guaranteed to meet at one point?

OR

Case study: A farmer's field ABCD has diagonals AC $= 70$ m and BD $= 70$ m which meet at right angles. A fence is built joining the midpoints P, Q, R, S of the sides in order. 5

(i) Name PQRS and justify.

(ii) Find each side.

(iii) Find the perimeter and the area of PQRS.

(iv) What is the area of the field?

39. (i) Find the distance between -4 and 3, and between $-\frac{5}{2}$ and $\frac{7}{2}$, on the number line. 5

(ii) Write $\left|-\frac{7}{9}\right|$, $|0|$ and $\left|\frac{5}{3}\right|$.

(iii) Between which two consecutive integers does $\frac{13}{4}$ lie?

OR

(i) Find the rational number x such that $x\left(\frac{1}{3} + \frac{1}{6}\right) = \frac{9}{2}$. 5

(ii) Verify $\frac{1}{2}\left(\frac{3}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{3}{4} + \frac{1}{2} \times \frac{1}{3}$.

40. A rectangular garden has length l m and width w m. Wire fencing costs ₹100/m along the length (both sides), wooden fencing ₹80/m along the width (both sides) and seeding ₹50/m^2. 5

(i) Write the expression for the total cost.

(ii) Find it for $l = 20$, $w = 10$.

(iii) Name the terms, variables and coefficients.

OR

Case study: A farmer cuts a 300-ft fence into two pieces; the longer is four times the shorter. A rectangular plot has length 3 more than twice its width and perimeter 24 m. 5

(i) Find the two pieces.

(ii) Find the dimensions of the plot.

(iii) Find the area of the plot.

END OF QUESTION PAPER **Total: 80 Marks**